

SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 07: Multivariate Analysis

LECTURE W7L1: Principal Component Analysis

BY

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CONCEPTS COVERED

Introduction to Multivariate Analysis and Principal Component Analysis

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Multivariate Analysis

- Many high-dimensional datasets are comprising of multiple variables. They are associated with each other, and such associations are complex.
- Multivariate analysis refers to a collection of methods used to untangle such complex, overlapping information and associations among these variables.
- It may be linear or nonlinear: In case of linear multivariate analysis, objective is to find out the linear combinations of the associated variables with a specific purpose, such as dimensionality reduction.
- Let $X_1, X_2, \ldots, X_p$ be the dependent variables and Z be the predictand, then,

$$Z = a_1X_1 + a_2X_2 + \cdots + a_pX_p = a'X$$

Therefore, $a_1, a_2, \ldots, a_p$ are the coefficients.

Methods in Multivariate Analysis

Multivariate Analysis contains a set of methods, such as :

a. Principal Component Analysis (PCA)

b. Supervised Principal Component Analysis (SPCA)

c. Canonical Correlational Analysis (CCA)

d. Empirical Orthogonal Function (EOF)

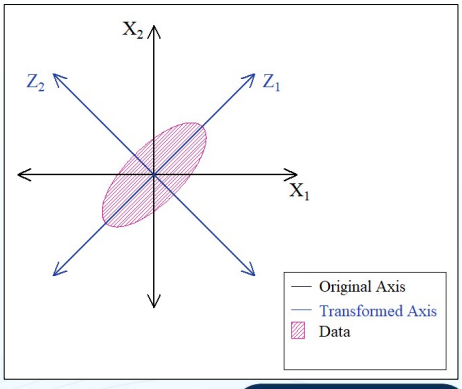
e. One-way and Two-way analysis of variance (ANOVA)

Principal Component Analysis

- Principal Component Analysis (PCA) is the transformation of p correlated variables into p uncorrelated orthogonal components through their linear combination.
- The resulting uncorrelated orthogonal components are known as Principal Components (PCs).
- In general, the objective of the PCA is data compression, in such a way that the resulting PCs are uncorrelated to each other.
- Consequently, the total variance of the original data is redistributed.
- The 1st PC contains maximum amount of variance and variance gradually decreases for the subsequent PCs.

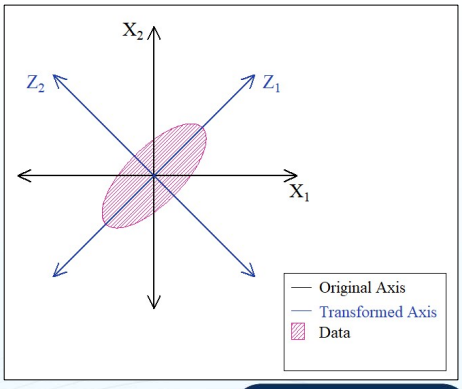
Principal Components Analysis

- The PCA reduces the dimensionality as the first few components explain most of the variance of the original data set.
- The PCA for p variables tries to reorient p^{th} -dimensional space (or Cartesian orthogonal coordinate system) to satisfy the redistribution of variance.
- The transformation of axes is geometrically illustrated for two variables in the figure.



Principal Components Analysis

- The original two variables X_1 and X_2 are transformed to Z_1 and Z_2.
- The first PC (Z_1) contains maximum amount of variance of the original data and gradually decreases for the subsequent components.
- Due to the orthogonality of the transformed axes (i.e., $Z_1, Z_2, \ldots, Z_p$), the correlation between them becomes zero.

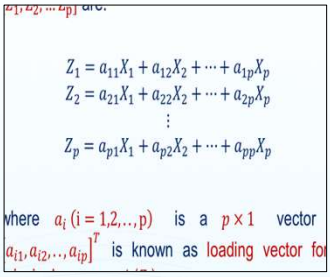


Derivation of Principal Components

- For a set of p variables, $X = [X_1, X_2, \ldots X_p]$ each having n observations, the set of principal components $[Z_1, Z_2, \ldots Z_p]$ are:

$$
\begin{aligned}
Z_1 &= a_{11}X_1 + a_{12}X_2 + \cdots + a_{1p}X_p \\
Z_2 &= a_{21}X_1 + a_{22}X_2 + \cdots + a_{2p}X_p \\
&\vdots \\
Z_p &= a_{p1}X_1 + a_{p2}X_2 + \cdots + a_{pp}X_p
\end{aligned}
$$

where $a_i\ (\mathrm{i} = 1,2,..,\mathrm{p})$ is a $p \times 1$ vector and $[a_{i1}, a_{i2}, .., a_{ip}]^T$ is known as loading vector for i^{th} principal component (Z_i).



Derivation of Principal Components: Loading Vectors

- Geometrically, the loading vector shows the direction of orientation of the PC axis.
- Therefore, the sum of squared terms given by $a_i^T a_i$ is 1.
- If $U = [a_1, a_2, .. a_p]$ is the orthogonal projection matrix, the PC matrix can be expressed as $Z = XU$.
- The PCs being uncorrelated, the covariance of any two PCs is zero.
- The variance of the PCs can be calculated as,

$$Var(Z_i) = Var(Xa_i) = a_i^T Cov(X) a_i = a_i^T S_x a_i$$

where S_x represents the covariance matrix of X.

Derivation of Principal Components

- The PCs can be determined by maximizing the variance of ith principal component with the constraint that sum of square of loadings is unity.
- The optimization problem can be expressed as,

maximize $Var(Z_i)$ or maximize $a_i^T S_x a_i$

subjected to $a_i^T a_i = 1$

- Further, the estimated i^{th} and j^{th} principal component should be such that $a_i^T S_x a_i \geq a_j^T S_x a_j$ for $i < j$.

Derivation of Objective Function

- The optimization problem can be solved by using the method of Lagrange multiplier.
- Method of Lagrange multiplier is an optimization technique to find maxima/minima for a function subject to equality constraint.
- If the optimization problem is expressed as

$$\text{maximize } f(x)$$

$$\text{subjected to } g(x) = 0$$

- The Lagrangian function with Lagrange multiplier λ is given by

$$\mathcal{L}(x, \lambda) = f(x) - \lambda g(x)$$

Optimization of the Objective Function

- The solution of the optimization problem is given by

$$\frac{d\mathcal{L}}{dx} = 0$$

- Hence, the solution is given by

$$\mathcal{L} = a_i^T S_x a_i - \lambda(a_i^T a_i - 1)$$

$$\frac{d\mathcal{L}}{da_i} = 0$$

$$\text{or, } (S_x - \lambda I)a_i = 0$$

- The characteristic equation is obtained by taking the determinant of the above equation.

$$|S_x - \lambda I| = 0$$

Determination of Principal Components

- The roots obtained from the $|S_x - \lambda I| = 0$ equation (λ_i, for $i \in \{1,2,\dots,p\}$ and $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$) are eigen values of covariance matrix of X.
- The variance of ith principal component is more compared to j^{th} principal component if $i < j$.
- The principal component loading vector for i^{th} principal component (a_i) is given by

$$(S_x - \lambda_i I)a_i = 0 \qquad \text{Such that } a_i^T a_i = 1$$

- Thus, the principal component is obtained using the following expression:

$$Z_i = Xa_i$$

where Z_i is the principal component with variance λ_i.

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Determination of Principal Components

- According to the properties of covariance matrix, the trace of the matrix (sum of diagonal elements) is the total variance of all the variables in X $(X_1, X_2, \dots, X_p)$.
- Also, from the properties of matrix, sum of all eigen values is equal to the trace of matrix.
- Hence, the variance explained by i^{th} principal component is given by

$$Z_i = \frac{\lambda_i}{\sum_{j=1}^{p} \lambda_j} = \frac{\lambda_i}{tr(S_x)}$$

- A similar expression can be obtained for the principal components from the correlation matrix.
- However, the principal components obtained from the correlation and covariance matrix are not same.

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Example

During the training of a Neural Network model, three input variables are used to predict a target variable. In order to check whether the dimensionality of inputs can be reduced, PCA is to be undertaken. The covariance matrix (A) of the input variables is as follows:

$$A = \begin{bmatrix} 8.23 & 2.52 & 1.24 \\ 2.52 & 5.42 & 0.62 \\ 1.24 & 0.62 & 1.16 \end{bmatrix}$$

Compute the loading vectors and principal components.

Solution

The characteristic equation to find the roots of the given matrix is:

$$(A - \lambda I) = 0$$

$$\begin{bmatrix} 8.23 - \lambda & 2.52 & 1.24 \\ 2.52 & 5.42 - \lambda & 0.62 \\ 1.24 & 0.62 & 1.16 - \lambda \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 0$$

The characteristic root equation is,

$$-\lambda^3 + 14.81\lambda^2 - 52.1682\lambda + 36.81 = 0$$

Therefore, eigenvalues are obtained as follows,

$$\lambda_1 = 9.928, \quad \lambda_2 = 3.943, \quad \lambda_3 = 0.939$$

Total variance is $9.928 + 3.943 + 0.939 = 14.81$

Solution

Percentage of variance explained by each eigenvalue:

- Percentage of variance explained by $\lambda_1 = \frac{9.928}{14.81} \times 100 = 67.04\%$.
- Percentage of variance explained by $\lambda_2 = \frac{3.943}{14.81} \times 100 = 26.62\%$.
- Percentage of variance explained by $\lambda_3 = \frac{0.939}{14.81} \times 100 = 6.34\%$.

Therefore, the calculation of eigenvectors corresponding to the eigenvalue is through solving the system of equations with three unknowns given three equations.

Solution

- Solving for the eigenvector corresponding to the first eigenvalue, $\lambda_1 = 9.928$

$$\begin{bmatrix} 8.23 - 9.928 & 2.52 & 1.24 \\ 2.52 & 5.42 - 9.928 & 0.62 \\ 1.24 & 0.62 & 1.16 - 9.928 \end{bmatrix} \begin{bmatrix} a_{11} \\ a_{12} \\ a_{13} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1.698 & 2.52 & 1.24 \\ 2.52 & -4.508 & 0.62 \\ 1.24 & 0.62 & -8.768 \end{bmatrix} \begin{bmatrix} a_{11} \\ a_{12} \\ a_{13} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- Since this is a homogeneous system, the solution will not be a single unique value. Therefore, assume $a_{13} = 1$.
- Then, the first two equations become

$$-1.698a_{11} + 2.52a_{12} + 1.24(1) = 0$$
$$2.52a_{11} - 4.508a_{12} + 0.62(1) = 0$$

Solution

- On rearranging and calculating for a_{11} and a_{12} we get

$$a_{11} = 5.548 \quad a_{12} = 3.203 \quad a_{13} = 1$$

- The eigenvector is proportional to

$$\begin{bmatrix} a_{11} \\ a_{12} \\ a_{13} \end{bmatrix} = \begin{bmatrix} 5.484 \\ 3.203 \\ 1 \end{bmatrix}$$

- Normalizing the vector,

$$\Rightarrow \sqrt{(5.484)^2 + (3.203)^2 + 1^2} = \sqrt{30.07 + 10.26 + 1} = \sqrt{41.33} = 6.43$$

$$\Rightarrow a_{11} = \frac{5.484}{6.43} = 0.853; \quad a_{12} = \frac{3.203}{6.43} = 0.498; \quad a_{13} = \frac{1}{6.43} = 0.155$$

- Thus the normalized vector is, $a_1 = \begin{bmatrix} 0.853 \\ 0.498 \\ 0.155 \end{bmatrix}$

Solution

- Similarly, the corresponding eigenvectors for the other two eigenvalues are also calculated. Thus, all the loading vectors are:

$$a_1 = \begin{bmatrix} 0.853 \\ 0.498 \\ 0.155 \end{bmatrix} \quad a_2 = \begin{bmatrix} 0.500 \\ -0.865 \\ 0.029 \end{bmatrix} \quad a_3 = \begin{bmatrix} -0.150 \\ -0.050 \\ 0.987 \end{bmatrix}$$

- Thus, the Principal Components are as follows:

$$Z_1 = 0.853X_1 + 0.498X_2 + 0.155X_3$$
$$Z_2 = 0.500X_1 - 0.865X_2 + 0.029X_3$$
$$Z_3 = -0.150X_1 - 0.050X_2 + 0.987X_3$$

Solution

Some notes:

- The first loading vector explains the maximum variance (67.04%) because it corresponds to the largest eigenvalue ($\lambda_1 = 9.928$).
- The dimensionality reduction is useful in this context because the first two principal components capture a large portion of the feature variability accounting for $67.04\% + 26.62\% = 93.66\%$ of the total variance.
- Thus, the original three-dimensional input feature space can be represented using two principal components with minimal information loss.





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Statistical and Machine Learning Methods: Basics and Applications

MODULE 07: Multivariate Analysis

LECTURE W7L2: Selection of Principal Components

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CONCEPTS COVERED

Selection of minimum number of principal components

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Dimensionality Reduction using PCA

- PCA can be used for dimensionality reduction by selecting a threshold for explained variance.
- The minimum number of principal components are selected that explain the threshold variance.
- The first few PCs are enough to explain most of the variance required for analysis.

Techniques to select the required number of PCs

i. **Total variance explained criteria:**

- A threshold cumulative percentage of total variance can be selected (say V_T), depending on the prediction problem and accuracy of the data measurement.
- The first k PCs are selected if they are able to explain at least the threshold amount of total variance, as expressed below:

  Select the minimum value of k for which,

$$\frac{\sum_{j=1}^{k} \lambda_j}{\sum_{i=1}^{p} \lambda_i} \geq V_T$$

Techniques to select the required number of PCs

ii. Average eigen value criteria:

The first k PCs corresponding to which the eigen value is above the mean eigenvalue are selected.

Select k PCs, if $\lambda_k \geq \bar{\lambda}$ and $\lambda_{k+1} < \bar{\lambda}$

iii. Scree plot:

- The plot between PCs and variance explained by each PC is called the scree plot.
- The first i^{th} PCs are selected for which scree plot shows significant slope.
- Based on the scree plot, a hypothesis test can be used for selecting PCs.

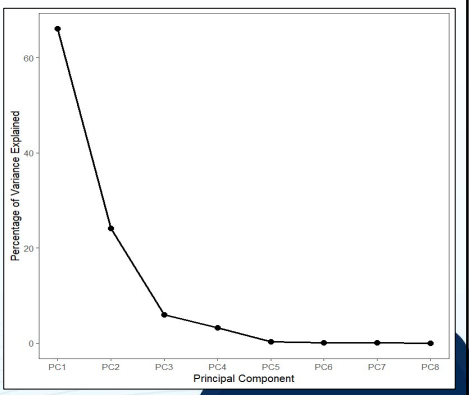
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Techniques to select the required number of PCs

- If k is the number of selected PCs, then the null hypothesis is the equality of all remaining eigen vectors (as they measure variance explained by any PC).

$$D = n\left[(p-k)ln\left(\overline{\lambda_k}\right) - \sum_{j=k+1}^{p} \ln\left(\lambda_j\right)\right]$$

where $\overline{\lambda_k} = \frac{\sum_{j=k+1}^{p} \lambda_j}{p-k}$



- The test statistics D follows χ^2 distribution with $0.5(p-k-1)(p-k+2)$ degrees of freedom.
- For a hypothesis test at α level of significance, the null hypothesis is rejected if $D > \chi^2_{\alpha}\left(0.5(p-k-1)(p-k+2)\right)$.

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Example-1

An artificial neural network (ANN) model is being trained using six input predictor features to predict one target variable. Before training the model, PCA is applied to reduce the dimensionality. The variance explained by each PC is given in table. Determine the minimum number of principal components required to retain at least 95% of the total variance in the training data.

Principal Component	Variance Explained
1st	0.571
2nd	0.272
3rd	0.106
4th	0.033
5th	0.012
6th	0.006

Solution

Principal Component	Variance Explained	Cumulative Variance Explained
1st	0.571	0.571
2nd	0.272	0.843
3rd	0.106	0.949
4th	0.033	0.982
5th	0.012	0.994
6th	0.006	1.000

- The last two principal components were insignificant and explained only explaining 0.3% of the total variance.
- For explaining 95% of the variance, the first four principal components are enough to train the ANN.

Example-2

To train a deep learning model, a dataset having 40 observations and 8 input variables is selected. Before training the model, PCA is applied to examine whether some of the input variables contain redundant information. The eigenvalues of the covariance matrix of the 8 input variables are: 20.75, 13.88, 7.57, 1.07, 1.02, 0.93, 0.87, and 0.71.

Test whether the last five eigenvalues differ significantly from one another at 5% level of significance.

Solution

Given, n = 40

Eigenvalues: $\lambda_1 = 20.75,\ \lambda_2 = 13.88,\ \lambda_3 = 7.57,\ \lambda_4 = 1.07,\ \lambda_5 = 1.02,\ \lambda_6 = 0.93,\ \lambda_7 = 0.87$ and $\lambda_8 = 0.71$.

- *Null Hypothesis:* Last five eigenvalues are equal, i.e., $\lambda_4 = \lambda_5 = \lambda_6 = \lambda_7 = \lambda_8$.
- *Alternative Hypothesis:* At least one eigenvalue out of the last five is not equal to the others.
- *Level of significance:* $\alpha = 5\%$

Solution

The test statistic is given by, $D = n\left[(p-k)\ln(\bar{\lambda}_k) - \sum_{j=k+1}^{p}\ln(\lambda_j)\right]$

$p = 8$ and $k = 8 - 5 = 3$.

$$\overline{\lambda_3} = \frac{\sum_{j=k+1}^{p}\lambda_j}{p-k} = \frac{\sum_{j=4}^{8}\lambda_j}{8-3} = \frac{1.07 + 1.02 + 0.93 + 0.87 + 0.71}{5} = 0.92$$

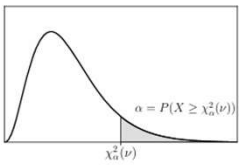
$$\sum_{j=k+1}^{p}\ln(\lambda_j) = \sum_{j=4}^{8}\ln(\lambda_j) = -0.47$$

$$D = 40\left[(8-3)\ln(\overline{\lambda_3}) - \sum_{j=4}^{8} ln(\lambda_j)\right] = 40(5\ln(0.92) + 0.47)$$

$$\boldsymbol{D = 2.12}$$

Solution

χ^2 Distribution Table



	α									
ν	0.999	0.990	0.950	0.900	0.500	0.100	0.050	0.010	0.005	0.001
1	0.000	0.000	0.004	0.016	0.455	2.706	3.841	6.635	7.879	10.828
2	0.002	0.020	0.103	0.211	1.386	4.605	5.991	9.210	10.597	13.816
3	0.024	0.115	0.352	0.584	2.366	6.251	7.815	11.345	12.838	16.266
4	0.091	0.297	0.711	1.064	3.357	7.779	9.488	13.277	14.860	18.467
5	0.210	0.554	1.145	1.610	4.351	9.236	11.070	15.086	16.750	20.515
6	0.381	0.872	1.635	2.204	5.348	10.645	12.592	16.812	18.548	22.458
7	0.598	1.239	2.167	2.833	6.346	12.017	14.067	18.475	20.278	24.322
8	0.857	1.646	2.733	3.490	7.344	13.362	15.507	20.090	21.955	26.125
9	1.152	2.088	3.325	4.168	8.343	14.684	16.919	21.666	23.589	27.877
10	1.479	2.558	3.940	4.865	9.342	15.987	18.307	23.209	25.188	29.588
11	1.834	3.053	4.575	5.578	10.341	17.275	19.675	24.725	26.757	31.264
12	2.214	3.571	5.226	6.304	11.340	18.549	21.026	26.217	28.300	32.910
13	2.617	4.107	5.892	7.042	12.340	19.812	22.362	27.688	29.819	34.528
14	3.041	4.660	6.571	7.790	13.339	21.064	23.685	29.141	31.319	36.123
15	3.483	5.229	7.261	8.547	14.339	22.307	24.996	30.578	32.801	37.697
16	3.942	5.812	7.962	9.312	15.338	23.542	26.296	32.000	34.267	39.252

Solution

The test statistics D is supposed to follow χ^2 distribution with $0.5(p-k-1)(p-k+2) = 0.5(5-1)(5+2) = 14$ degrees of freedom.

$\chi^2_\alpha(14) = 23.68$ (from distribution table)

As $D < 23.68$ $\left(\chi^2_{0.05}(14)\right)$, null hypothesis of the eigenvalues being equal cannot be rejected at 5% significance level.

The last five eigenvalues are not significantly different, indicating that they have less informative variation.

Hence, instead of using all 8 input variables, the deep learning model may be trained using only the first three principal components, which explain most of the total variance.





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LECTURE W7L3 : Empirical Orthogonal Function

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CONCEPTS COVERED

Multivariate Analysis: Empirical Orthogonal Function

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Empirical Orthogonal Function

- Real-world data, apart from time variability, also exhibit spatial variation.
- PCA can also be utilized for studying these spatio-temporal variation in real-world data.
- PCA helps in understanding the contribution of each of the variables (X_i) to total variability (S_{xx}) (through the coefficient of loading vector).
- If the same variables across many locations are treated as different variables, then PCA can be used to study the relative contribution of different locations in total variability.
- This spatial analysis is known as the Empirical Orthogonal Function (EOF) analysis.

Empirical Orthogonal Function

- For EOF analysis, the dataset is collected in 3-D matrix.
- The first two dimensions show the grid location on which the variable is observed, and the last dimension shows the time steps.
- Hence, dataset for a variable has dimensions of $p_1 \times p_2 \times n$, where p_1 and p_2 are the number of stations or grid locations in x- and y- directions, respectively, and n is the number of time steps.
- First, the data is converted to $p \times n$ matrix, where $p = p_1 \times p_2$ by rearranging all the grid locations.

Covariance matrix for spatio-temporal data

- Let us consider the case of the monthly time series data, for instance, temperature data observed at p grid locations over a period of n months.
- The anomaly (deviation from mean) values are computed at each grid location by subtracting the corresponding mean of that grid location from the observed values.
- The anomaly values at each grid location can be represented by a vector, $X_i = (x_i(1), x_i(2), \ldots, x_i(n))$ and the anomaly data from all grid locations can be arranged into a matrix.

$$X = \begin{bmatrix} x_1(1) & x_1(2) & \ldots & x_1(n) \\ x_2(1) & x_2(2) & \cdots & x_2(n) \\ \vdots & \vdots & \ddots & \vdots \\ x_p(1) & x_p(2) & \cdots & x_p(n) \end{bmatrix}$$

where matrix X is of order $p \times n$.

Covariance matrix for spatio-temporal data

- The covariance matrix S is then computed from the mean-centered anomaly data as:

$$S = \frac{1}{n-1} XX^T$$

Since matrix X is of order $p \times n$, X^T will be $n \times p$. Therefore, XX^T has dimension $p \times p$.

- Hence, the resulting covariance matrix S is of dimension $p \times p$, where each element S_{ij} represents the covariance between the anomaly series at grid locations i and j.
- The PCA loadings and corresponding principal components are calculated using the same procedure followed for PCA.

Empirical Orthogonal Function

- The loading vector obtained for i^{th} principal component has p loadings, where p is the number of grid locations.
- If the square of the loading vector is arranged in $p_1 \times p_2$ matrix, it shows the relative contribution of the variable at a grid location to the spatial distribution of variance of ith principal component.
- The principal components obtained in the EOF analysis show the variability across the space.

Example

The five monthly mean values (n) at four monitoring agricultural fields (p) are shown in matrix A of order $p \times n$. Since the fields are close to one another, the data may be highly correlated. Calculate the variance explained and EOF loading vector corresponding to the largest eigenvalue.

$$A = \begin{bmatrix} 1.86 & 1.95 & 2.26 & 2.37 & 2.64 \\ 2.75 & 2.83 & 3.58 & 3.36 & 3.54 \\ 4.85 & 4.95 & 4.63 & 3.85 & 3.34 \\ 5.86 & 5.54 & 5.25 & 4.63 & 4.35 \end{bmatrix}$$

Solution

- Given: the five monthly mean values at four stations are recorded, n = 5
- Calculation of station-wise mean values:

$$\bar{A} = [2.216, 3.212, 4.324, 5.126]^T$$

- Calculation of station-wise anomalies (deviations from mean):

$$A - \bar{A} = X = \begin{bmatrix} -0.356 & -0.266 & 0.044 & 0.154 & 0.424 \\ -0.462 & -0.382 & 0.368 & 0.148 & 0.328 \\ 0.526 & 0.626 & 0.306 & -0.474 & -0.984 \\ 0.734 & 0.414 & 0.124 & -0.496 & -0.776 \end{bmatrix}$$

Solution

- The covariance matrix is calculated as:

$$S = \frac{1}{n-1} XX^T$$

$$S = \frac{1}{5-1} \begin{bmatrix} -0.356 & -0.266 & 0.044 & 0.154 & 0.424 \\ -0.462 & -0.382 & 0.368 & 0.148 & 0.328 \\ 0.526 & 0.626 & 0.306 & -0.474 & -0.984 \\ 0.734 & 0.414 & 0.124 & -0.496 & -0.776 \end{bmatrix} \begin{bmatrix} -0.356 & -0.462 & 0.526 & 0.734 \\ -0.266 & -0.382 & 0.626 & 0.414 \\ 0.044 & 0.368 & 0.306 & 0.124 \\ 0.154 & 0.148 & -0.474 & -0.496 \\ 0.424 & 0.328 & -0.984 & -0.776 \end{bmatrix}$$

$$S = \begin{bmatrix} 0.100 & 0.111 & -0.208 & -0.192 \\ 0.111 & 0.156 & -0.191 & -0.194 \\ -0.210 & -0.191 & 0.488 & 0.420 \\ -0.193 & -0.194 & 0.420 & 0.394 \end{bmatrix}$$

Solution

- Therefore, the eigenvalues calculated for the covariance matrix,

$$\lambda_1 = 1.540, \lambda_2 = 0.073, \lambda_3 = 0.012, \ \lambda_4 = 0$$

- The proportion of variance explained by any EOF is the ratio of corresponding eigenvalue and trace of covariance matrix.
- Thus, the proportion of variance explained by first eigenvalue is:

$$\lambda_1 = \frac{1.54}{1.540 + 0.073 + 0.012 + 0} = 0.955$$

- The EOF loading vector corresponding to $\lambda_1 = 1.54$, is obtained as:

$$a_1 = \begin{bmatrix} -0.304 \\ -0.310 \\ 0.666 \\ 0.605 \end{bmatrix}$$

Solution

- Similarly, the EOF loading vectors corresponding to other eigenvalues can be obtained.
- The first EOF loading vector corresponding to $\lambda_1 = 1.54$, explains 95.5% of the total variance in the four agricultural fields.
- It can be observed that the variance explained by the first EOF is more than 95%.


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Statistical and Machine Learning Methods: Basics and Applications

MODULE 07 : Multivariate Analysis

LECTURE W7L4: Analysis of Variance-I

BY

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CONCEPTS COVERED

Multivariate Analysis: One-Way Analysis of Variance

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Analysis of Variance

- Analysis of Variance (ANOVA) is a statistical procedure to check the significance of variance in different samples of the same population and thus check the null hypothesis, i.e., means of all samples are equal.
- ANOVA is also used to study the spatial homogeneity of datasets.
- Under ANOVA, variance is partitioned into a number of sources/factors.
- Depending upon the number of sources of variance (referred as attributes), apart from the system error (experimental or measurement errors), one-way ANOVA or two-way ANOVA is used.

One-Way Analysis of Variance

- One-Way ANOVA is used when only a single factor/attribute contributes to the variance.
- Due to this attribute, the sample mean differs from the population mean.
- Therefore, the sample mean can be written as the sum of the population mean and the effect of the attribute.

$$\bar{x_i} = \mu + \alpha_i$$

where $\bar{x_i}$ is sample mean of ith sample, μ is the mean of all the samples or population mean and α_i is effect of attribute on the sample mean.

One-Way Analysis of Variance

- In One-Way ANOVA, the null hypothesis is that all the sample means are equal to the population mean.

$$H_0: \overline{x_1} = \overline{x_2} = \cdots = \overline{x_a} = \mu$$

or, $H_0: \alpha_i = 0$ for all $i \in \{1,2,\ldots,a\}$

where $\alpha_i = \overline{x_i} - \mu$ for $i = 1,2,\ldots,a$.

- Correspondingly, the alternative hypothesis states that at least one sample mean is not equal to the population mean.

$H_1: \alpha_i \neq 0$ for at least one value for i

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One-Way Analysis of Variance

- Null hypothesis can only be true if all the variability is primarily contributed due to chance or random error.
- Hence, the variance in the dataset needs to be separated into variance due to attribute and variance due to random error.
- Ratio of mean variance contribution from these two categories is the test statistic for testing the null hypothesis.
- Therefore, an observation from i^{th} sample can be expressed as sum of population mean, attribute effect and random error.

$$x_{ij} = \overline{x_i} + e_{ij} = \mu + \alpha_i + e_{ij}$$

where e_{ij} is the random error associated with the element x_{ij}.

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One-Way Analysis of Variance

- Also, the equation can be re-written in terms of deviations from the sample or population mean as

$$x_{ij} - \mu = \alpha_i + e_{ij}$$

$$\text{or,} (x_{ij} - \mu) = (\bar{x}_i - \mu) + (x_{ij} - \bar{x}_i)$$

- If different samples are available and i^{th} sample contains n_i observations, the total sum of squares (SST) is defined as the sum of squared deviation of all observations from the population mean.

One-Way Analysis of Variance

- Therefore, the equation is obtained by squaring both sides of the previous equation and taking the summation for all the observations.

$$\sum_{i=1}^{a}\sum_{j=1}^{n_i}(x_{ij} - \mu)^2 = \sum_{i=1}^{a} n_i(\bar{x}_i - \mu)^2 + \sum_{i=1}^{a}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2$$

$$SST = SSA + SSE$$

where a is the number of different attributes/factors, n_i is the number of data in i^{th} sample, SSA (Sum Square due to Attribute) shows the variance contributed by attribute and SSE (Sum Square Error) shows the variance due to random error.

- Hence, the total variance is partitioned into variance due to attribute and variance due to random errors.

One-Way Analysis of Variance

- Degrees of freedom for each term can be evaluated as follows:
  - While calculating SSA, one mean μ is computed from the entire data.
  - So, one degree of freedom (dof) is lost. Therefore, degree of freedom for SSA is $a-1$.
  - Similarly, there is one mean for each attribute, counting total of a attributes' mean that are used to calculate SSE. And one mean μ is used to compute SST. Thus, dof of SSE and SST are $N-a$ and $N-1$ respectively.
- Using these degrees of freedom, the mean square error (MSE) and mean square attribute (MSA) can be calculated as,

$$MSE = \frac{SSE}{N-a}; \quad MSA = \frac{SSA}{a-1}$$

One-Way Analysis of Variance

- The test statistics (F) in one-way ANOVA is given by the ratio of MSA and MSE.

$$F = \frac{MSA}{MSE}$$

- A large value of test statistics (F) indicates that the effect of attribute is prominent as compared to effect of the random error over total variance and thus all the sample means are not equal.
  - The test statistics follows F-distribution with $(a-1)$ and $(N-a)$ degrees of freedom.
  - Hence, the null hypothesis is rejected if

    $F > F_\alpha(a-1, N-a)$ at α level of significance.

One-Way Analysis of Variance

- Sum of squares can also be obtained using following equations:

$$SST = \sum_{i=1}^{a}\sum_{j=1}^{n_i} x_{ij}^2 - C$$

$$SSA = \sum_{i=1}^{a} \frac{T_i^2}{n_i} - C$$

$$SSE = SST - SSA$$

$$C = \frac{(\sum_{i=1}^{a} T_i)^2}{N}$$

where T_i represents the sum of observations in i^{th} sample $(T_i = \sum_{j=1}^{n_i} x_{ij})$ and C is called correction term for the mean.

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One-Way Analysis of Variance

The details of the one-way ANOVA are summarized in the following table:

Source of variation	Degree of freedom	Sum of squares	Mean Square	F
Attribute	$a-1$	SSA	MSA	$\frac{MSA}{MSE}$
Error	$N-a$	SSE	MSE	
Total	$N-1$	SST		

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Example

The validation RMSE values obtained from five independent runs of Artificial Neural Network (ANN) models with 1, 2 and 3 hidden layers are presented in the table. Using one-way ANOVA, test whether the RMSE differs significantly among the ANN architectures at the 5% level of significance.

Number of Hidden layers	Run 1	Run 2	Run 3	Run 4	Run 5
1	5.26	5.85	6.36	7.24	8.42
2	5.36	5.94	6.85	7.68	8.76
3	5.84	6.25	6.98	7.9	8.98

Solution

- In the given dataset, the number of hidden layers is the only source of variation; hence, it can be treated as the attribute for ANOVA analysis.
- Denoting $x_i (i = 1,2,3)$ as RMSE from i^{th} hidden layer, μ as overall mean and $\alpha_i = \bar{x_i} - \mu$.
- *Null Hypothesis*: RMSE do not differ significantly. $\alpha_i = 0$ for $i \in \{1,2,3\}$.
- *Alternative Hypothesis*: $\alpha_i \neq 0$ for at least one value of i.
- *Level of Significance*: $\alpha = 5\%$.
- Also, from the given table, $N = 15$, $a = 3$.

Solution

- Different quantities of one-way ANOVA table can be calculated using the following equation:

$$\sum_{i=1}^{3}\sum_{j=1}^{5}(x_{ij}-\mu)^2 = \sum_{i=1}^{3}5(\bar{x_i}-\mu)^2 + \sum_{i=1}^{3}\sum_{j=1}^{5}(x_{ij}-\bar{x_i})^2$$

- Let X_1, X_2 and X_3 represent RMSE values corresponding to ANN models with 1, 2, and 3 hidden layers, respectively.
- The RMSE obtained from the jth run of the ANN model with i^{th} hidden-layer configuration is represented by $x_{i,j}$.

$$\text{Overall mean } (\mu) = \frac{\sum_{i=1}^{3}\sum_{j=1}^{5}x_{i,j}}{5\times 3} = 6.91$$

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Solution

- The RMSE for ANN models with one hidden layer is given by

$$\bar{x_1} = \frac{\sum_{j=1}^{5}x_{1,j}}{5} = \frac{5.26+5.85+6.36+7.24+8.42}{5} = 6.63$$

- Similarly,

$$\bar{x_2} = \frac{\sum_{j=1}^{5}x_{2,j}}{5} = \frac{5.36+5.94+6.85+7.68+8.76}{5} = 6.92$$

$$\bar{x_3} = \frac{\sum_{j=1}^{5}x_{3,j}}{5} = \frac{5.84+6.25+6.98+7.90+8.98}{5} = 7.20$$

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Solution

Therefore,

$$SST = \sum_{i=1}^{3}\sum_{j=1}^{5}\left(x_{i,j} - \mu\right)^2 = \sum_{i=1}^{3}\sum_{j=1}^{5}\left(x_{i,j} - 6.91\right)^2 = 20.75$$

$$SSA = \sum_{i=1}^{3} 5(\bar{x}_i - \mu)^2 = 5 \times ((6.63 - 6.91)^2 + (6.92 - 6.91)^2 + (7.19 - 6.91)^2) = 0.80$$

$$SSE = \sum_{i=1}^{3}\sum_{j=1}^{5}\left(x_{i,j} - \bar{x}_i\right)^2$$

$$= \sum_{j=1}^{5}\left(x_{1,j} - 6.3\right)^2 + \sum_{j=1}^{5}\left(x_{2,j} - 6.92\right)^2 + \sum_{j=1}^{5}\left(x_{3,j} - 7.2\right)^2$$

$$SSE = 19.95$$

Solution

- The degree of freedom for SST, SSA and SSE are $(N - a) = 15 - 1 = 14$, $(a - 1) = 3 - 1 = 2$, and $(N - a) = 15 - 3 = 12$ respectively.
- Hence,

$$MSA = \frac{SSA}{a - 1} = \frac{0.8}{2} = 0.4$$

$$MSE = \frac{SSE}{N - a} = \frac{19.95}{12} = 1.66$$

- Thus,

$$F = \frac{MSA}{MSE} = \frac{0.4}{1.66} = 0.241$$

Solution

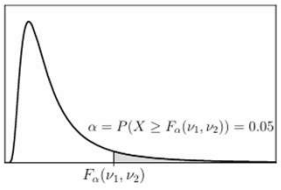
The details of one-way ANOVA for the given problem can be summarized in a table, as shown:

Source of variation	Degree of freedom	Sum of squares	Mean Square	F
Attribute	2	0.8	0.4	0.241
Error	12	19.95	1.66	
Total	14	20.75		

The test statistics (F) for ANOVA analysis follows F-distribution with $(a-1)=2$ and $(N-a)=12$ degrees of freedom.

Solution

F-distribution table $(\alpha = 0.05)$



ν_2 \ ν_1	1	2	3	4	5	6	7	8	9	10	12	15	20	30	50	60
1	161	199	216	224	230	234	237	239	241	242	244	246	248	250	252	252
2	18.5	19.0	19.2	19.2	19.3	19.3	19.3	19.4	19.4	19.4	19.4	19.4	19.4	19.5	19.5	19.5
3	10.1	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.62	8.58	8.57
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.75	5.70	5.69
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.50	4.44	4.43
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.81	3.75	3.74
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.38	3.32	3.30
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.08	3.02	3.01
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.86	2.80	2.79
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.70	2.64	2.62
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.57	2.51	2.49
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.47	2.40	2.38
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.38	2.31	2.30
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.31	2.24	2.22

Solution

- From the table, the value of, $F_{\alpha}\big((a-1),(N-a)\big)$ is obtained as: $F_{0.05}(2,12) = 3.89$
- Therefore, as $0.241 < 3.89$ ($F_{0.05}(2,12)$, null hypothesis cannot be rejected at the 5% significance level.
- Therefore, the RMSE does not differ significantly among ANN models with different numbers of hidden layers.





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Statistical and Machine Learning Methods: Basics and Applications

MODULE 07 : Multivariate Analysis

LECTURE W7L5: Analysis of Variance-II

BY

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CONCEPTS COVERED

Multivariate Analysis: Two-Way Analysis of Variance

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Two-Way Analysis of Variance

- Two-way ANOVA is used when two factors (instead of one) contribute to the variance, apart from the random error (white noise). These factors are known as called attribute-1 and attribute-2.
- Under this condition, the sample mean $(\bar{x}_i)$ for i^{th} attribute-1 and j^{th} attribute-2 is given by

$$\bar{x}_i = \mu + \alpha_i + \beta_j$$

where $\bar{x}_i$ is sample mean of ith sample, μ is the mean of all the samples or population mean, α_i and β_j are the effects of ith component of attribute-1 and j^{th} component of attribute-2 on sample mean.

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Two-Way Analysis of Variance

- In two-way ANOVA, the null hypothesis is that the effects across the different components of each attributes are same.

  Thus, $H_0: \alpha_i = 0$ and $\beta_j = 0$ for all $i \in \{1,2,\dots,a\}$ and $j \in \{1,2,\dots,b\}$.

  where a and b are the number of components of attribute-1 and attribute-2 respectively.
- Correspondingly, the alternative hypothesis states that at least one sample mean is not equal to the population mean.

H_a: Null hypothesis is not true

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Two-Way Analysis of Variance

- Null hypothesis can only be true if all the variabilities are primarily contributed due to chance or random error.
- Hence, the variance in the dataset need to be separated into variance due to components of attribute-1, variance due to components of attribute-2, and variance due to random error.
- Therefore, using the equation $\bar{x}_i = \mu + \alpha_i + \beta_j$, an observation from i^{th} sample is expressed as,

$$x_{ij} = \bar{x}_i + e_{ij} = \mu + \alpha_i + \beta_j + e_{ij}$$

where e_{ij} is the random error associated with element x_{ij}.

Two-Way Analysis of Variance

- Thus, the total variance can be expressed as:

$$\sum_{i=1}^{a}\sum_{j=1}^{b}(x_{ij} - \bar{x})^2 = b\sum_{i=1}^{a}(\bar{x}_i - \bar{x})^2 + a\sum_{j=1}^{b}(\bar{x}_j - \bar{x})^2 + \sum_{i=1}^{a}\sum_{j=1}^{b}(x_{ij} - \bar{x}_i - \bar{x}_j + \bar{x})^2$$

$$SST = SSA_1 + SSA_2 + SSE$$

where $\bar{x}$ is overall mean, $\bar{x}_i$ is mean of all the observations for i^{th} component of attribute-1, $\bar{x}_j$ is the mean of all the observations for j^{th} component of attribute-2, a is the number of components in attribute-1 and b is the number of components in attribute-2.

- SSA_1 and SSA_2 represent the variance contributed due to the effect of attribute-1 and attribute-2, respectively.

Two-Way Analysis of Variance

- The degrees of freedom for SSA_1, SSA_2 and SSE are $(a-1)$, $(b-1)$, and $(a-1)(b-1)$ respectively.
- The degrees of freedom for SST is $ab-1$.
- Using the degrees of freedom, the mean of SSA_1, SSA_2 and SSE (i.e., MSA_1, MSA_2 and MSE) are calculated.
- The test statistics for two-way ANOVA is defined by,

$$F_1 = \frac{MSA_1}{MSE}; \qquad F_2 = \frac{MSA_2}{MSE}$$

where, F_1 follows the F-distribution with $(a-1)$ and $(a-1)(b-1)$ degrees of freedom. Similarly, F_2 follows F-distribution with $(b-1)$ and $(a-1)(b-1)$ degrees of freedom.

Two-Way Analysis of Variance

Null Hypothesis of no significant difference in mean cannot be rejected at α significance level if,

$F_1 > F_\alpha((a-1),(a-1)(b-1))$ and $F_2 > F_\alpha((b-1),(a-1)(b-1))$

Details of two-way ANOVA are summarized in the Table.

Source of variation	Degree of freedom	Sum of squares	Mean Square	F
Attribute-1	$a-1$	SSA_1	MSA_1	$\frac{MSA_1}{MSE}$
Attribute-2	$b-1$	SSA_2	MSA_2	$\frac{MSA_2}{MSE}$
Error	$(a-1)(b-1)$	SSE	MSE	
Total	$N-1$	SST		

Two-Way Analysis of Variance

Different partitions of variance can also be calculated using the following relationships:

$$SST = \sum_{i=1}^{a}\sum_{j=1}^{b} x_{ij}^2 - C$$

$$SSA_1 = \frac{1}{b}\sum_{i=1}^{a} T_{i\bullet}^2 - C$$

$$SSA_2 = \frac{1}{a}\sum_{j=1}^{b} T_{\bullet j}^2 - C$$

Two-Way Analysis of Variance

- Also, the SSE is calculated as follows:

$$SSE = SST - SSA_1 - SSA_2$$

$$C = \frac{\left(\sum_{i=1}^{a}\sum_{j=1}^{b} x_{ij}\right)^2}{ab}$$

where $T_{i\bullet}$ and $T_{\bullet j}$ represents the sum of observations for i^{th} component of attribute-1 and sum of observations for j^{th} component of attribute-2, respectively, and C is called correction term for the observation mean.

Example

It is required to investigate the effect of different models (M) and input datasets (I) on the performance of a regression task. The following table presents the regression accuracy (%) achieved on the test dataset for different combinations of models (M) and input datasets (I).

Check whether the regression accuracy differs significantly either due to models (M) or input datasets (I) selected at 1% level of significance.

Models	Input datasets		
	$I1$	$I2$	$I3$
$M1$	90.0	87.5	96.5
$M2$	96.0	92.0	95.5
$M3$	95.0	97.5	93.5
$M4$	87.0	79.5	83.5

Solution

- There are two sources of variance, one being models (M) and other being input datasets (I).
- Different types of models (M) and different inputs (I) can be considered as first and second attributes.
- Hence, $a = 4$ and $b = 3$.
- *Null Hypothesis*: The regression accuracy does not differ significantly due to different models (M) or input datasets (I) selected i.e., $\alpha_i = 0$ for $i \in \{1,2,3,4\}$ and $\beta_j = 0$ for $j \in \{1,2,3\}$.
- *Alternate Hypothesis*: Null hypothesis is not true.
- *Level of Significance*: $\alpha = 1\%$.

Solution

Different test statistics for two-way ANOVA can be calculated as follows:

The sum of observations for i^{th} component of attribute-1 are obtained as:

$$T_{1\bullet} = (90.0 + 87.5 + 96.5) = 274.0$$

Similarly, $T_{2\bullet} = 283.5$, $T_{3\bullet} = 290.0$, and $T_{4\bullet} = 250.0$

The sum of observations for j^{th} component of attribute-2 are obtained as:

$$T_{\bullet 1} = (90.0 + 96.0 + 95.0 + 87.0) = 368.0$$

Similarly, $T_{\bullet 2} = 356.5$, and $T_{\bullet 3} = 373.0$

Solution

Correction term (C) is calculated as,

$$C = \frac{\left(\sum_{i=1}^{4}\sum_{j=1}^{3} x_{ij}\right)^2}{ab} = \frac{1204506}{12} = 100375.50$$

$$SST = \sum_{i=1}^{a}\sum_{j=1}^{b} x_{ij}^2 - C = 392.23$$

$$SSA_1 = \frac{1}{b}\sum_{i=1}^{a} T_{i\bullet}^2 - C = \frac{302048.30}{3} - 100375.50 = 307.230$$

$$SSA_2 = \frac{1}{a}\sum_{j=1}^{b} T_{\bullet j}^2 - C = \frac{401645.30}{4} - 100375.50 = 35.80$$

$$SSE = SST - SSA_1 - SSA_2 = 49.21$$

Solution

- The mean square sum (MSA_1, MSA_2 and MSE) can be calculated by dividing respective sum of squares (SSA_1, SSA_2 and SSE) with their respective degrees of freedom.
- The degrees of freedom for SSA_1, SSA_2 and SSE are $(a-1)=3$, $(b-1)=2$ and $(a-1)(b-1)=6$ respectively.
- Thus,

$$MSA_1 = \frac{SSA_1}{a-1} = \frac{307.23}{3} = 102.41$$

$$MSA_2 = \frac{SSA_2}{b-1} = \frac{35.8}{2} = 17.90$$

$$MSE = \frac{SSE}{(a-1)(b-1)} = \frac{49.21}{6} = 8.20$$

Solution

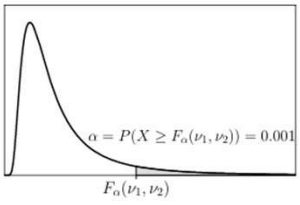
- Therefore, $F_1 = \frac{MSA_1}{MSE} = \frac{102.41}{8.20} = 12.50$ and $F_2 = \frac{MSA_2}{MSE} = \frac{17.9}{8.20} = 2.20$
- All the calculated values are summarized in the following two-way ANOVA table:

Source of variation	Degree of freedom	Sum of squares	Mean Square	*F*
Attribute-1	3	307.23	102.41	12.50
Attribute-2	2	35.80	17.90	2.20
Error	6	49.21	8.20	
Total	11	392.23		

- The test statistics F_1 is supposed to follow F-distribution (table in next slide) with $(a-1)$ and $(a-1)(b-1)$, i.e., 3 and 6 degrees of freedom.

Solution

F-distribution table at $\alpha = 0.01$



$\nu_1 \longrightarrow$

ν_2	1	2	3	4	5	6	7	8	9	10	12	15	20	30	50	60
1	4052	4999	5403	5624	5763	5859	5928	5981	6022	6056	6106	6157	6209	6261	6302	6313
2	98.5	99.0	99.2	99.2	99.3	99.3	99.4	99.4	99.4	99.4	99.4	99.4	99.4	99.5	99.5	99.5
3	34.1	30.8	29.5	28.7	28.2	27.9	27.7	27.5	27.4	27.2	27.0	26.9	26.7	26.5	26.3	26.3
4	21.2	18.0	16.7	16.0	15.5	15.2	15.0	14.8	14.7	14.5	14.4	14.2	14.0	13.8	13.7	13.6
5	16.3	13.3	12.1	11.4	11.0	10.7	10.5	10.3	10.2	10.0	9.89	9.72	9.55	9.38	9.24	9.20
6	13.8	10.9	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.72	7.56	7.40	7.23	7.09	7.06
7	12.2	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.47	6.31	6.16	5.99	5.86	5.82
8	11.3	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.67	5.52	5.36	5.20	5.07	5.03
9	10.6	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.11	4.96	4.81	4.65	4.52	4.48
10	10.0	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.71	4.56	4.41	4.25	4.12	4.08

Solution

- From the table, $F_\alpha\big((a-1),(a-1)(b-1)\big)$ is obtained as: $F_{0.01}(3,6) = 9.78$
- The test statistics F_2 is supposed to follow F-distribution with $(b-1)$ and $(a-1)(b-1)$, i.e., 2 and 6 degrees of freedoms.
- Thus, $F_\alpha\big((b-1),(a-1)(b-1)\big)$ is obtained as: $F_{0.01}(2,6) = 10.92$
- Since $12.50\ (F_1) > 9.78$, it indicates that there is significant difference between the different models (M). However, $2.2\ (F_2) < 10.92$, it indicates that there is no significant difference between the input datasets (I) at 1% significance level.


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Thank You!